FAIR Alignment and Research Infrastructure as Code I

The template's design aligns with both the original FAIR principles [@wilkinson2016fair] and the FAIR for Research Software (FAIR4RS) principles [@barker2022fair4rs] at the repository level. FAIR4RS recognizes that software has requirements distinct from data—executability, composability, and dependency management—and the template addresses each.

FAIR Alignment and Research Infrastructure as Code II

Principle-by-Principle Alignment

Findability. Outputs are *Findable* through standardized directory structures, manifest files, and machine-readable metadata embedded in PDFs. Every project's `config.yaml` provides structured metadata (title, authors, DOIs, keywords) in a format parseable by both Pandoc and external indexing services. The `metrics.json` output provides a machine-readable inventory of all generated artifacts, their locations, and their provenance hashes.

Accessibility. Outputs are *Accessible* via open-source distribution on GitHub, with metadata embedded in the artifact itself rather than in a separate registry. The steganographic layer embeds provenance information directly in the PDF—including SHA-256 content hashes, build timestamps, and QR-encoded metadata—ensuring accessibility even when the PDF circulates outside the repository.

Interoperability. *Interoperability* is achieved through standard formats (PDF, JSON, BibTeX, YAML) and well-defined module APIs that enable cross-project composition. The Pandoc rendering pipeline accepts any Markdown with LaTeX input conforming to